Roadmap. Srednicki §6: the path integral for quantum mechanics. Time slicing gives $\langle q'', t''|q', t'\rangle = \int \mathcal{D}p\,\mathcal{D}q\, e^{i\int dt\,(p\dot{q}-H)}$, then the Lagrangian form; operator insertions become time-ordered products generated by sources; and $H \to (1-i\epsilon)H$ projects onto the vacuum, leaving $\langle 0|0\rangle_f$. LecNote6 (§7) evaluates it for the harmonic oscillator; LecNote7–8 (§§8–9) carry it to fields. Supplements: FeynmanPropagator (what the $i\epsilon$ does), WeylOrdering (p. 4).

Path Integral. Sec. 6th

Consider a non-relativistic QM in one dimension.

$$H(P,Q) = \frac{1}{2m}P^2 + V(Q).$$

P and Q are operators. $[Q,P] = i\hbar$.

We want to evaluate the particle transition probability.

Starts at q' at t' and end at q'', t''

$$\langle q''|e^{-\frac{i}{\hbar}Ht''}\, e^{\frac{i}{\hbar}Ht'}|q'\rangle.$$

where $|q''\rangle$ and $|q'\rangle$ are eigenstates of Q.

as the time evolution of $|q'\rangle$ is $e^{\frac{i}{\hbar}Ht}$

formulating this question in the Heisenberg picture [1]

$$Q(t) = e^{\frac{i}{\hbar}Ht}\, Q(0)\, e^{-\frac{i}{\hbar}Ht}$$

[1] Not needed until p. 6: pp. 2–5 use only $\langle q''|e^{-iH(t''-t')}|q'\rangle$. $Q(t) = e^{iHt}Qe^{-iHt}$ and $|q,t\rangle = e^{iHt}|q\rangle$ are used from p. 6 on, for the time-ordered insertions that appear in LSZ.

The instantaneous eigenstate of $Q(t)$ can be defined via

$$Q(t)\,|q,t\rangle = q\,|q,t\rangle$$

and $$|q,t\rangle = e^{+\frac{i}{\hbar}Ht}\,|q\rangle$$

where $$Q|q\rangle = q|q\rangle$$

$t=0$.

Then the transition amplitude is as simple as

$\langle q'', t'' \mid q', t' \rangle$ in the Heisenberg picture.

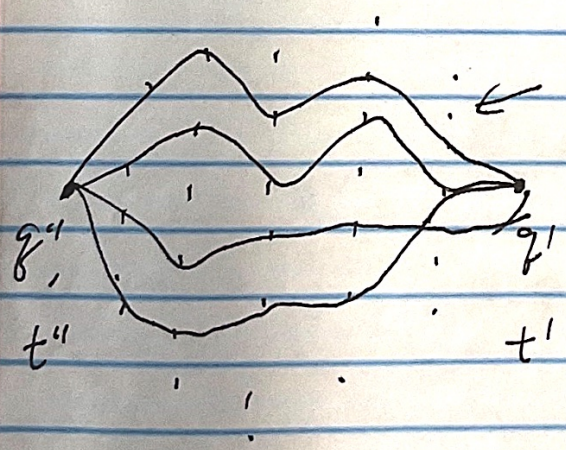
To evaluate, we can do the following.

$$\langle q'', t'' \mid q', t' \rangle = \int \prod_{j=1}^{N} dq_j \cdot \langle q'' | e^{-iH\delta t} | q_N \rangle \langle q_N |$$

$$e^{-iH\delta t} | q_{N-1} \rangle \langle q_{N-1} | e^{-iH\delta t} | \cdots \cdots \langle q_1 | e^{-iH\delta t} | q' \rangle$$

by inserting N complete states. $|q_i\rangle\langle q_i| \int_{-\infty}^{\infty} dq_i$

basically I am evaluating the transition as a product of small ~~t~~ transitions $\delta t = \frac{t'' - t'}{N}$ (1)



← each path has different weight (2)

1 N insertions give $N+1$ factors of $e^{-iH\delta t}$, so $\delta t = (t'' - t')/(N+1)$, matching the $N+1$ momentum integrals on p. 4. Irrelevant as $N \to \infty$.

2 Every path has weight $e^{iS[q]}$ of modulus one: paths differ in phase, not size. Near the classical path S is stationary and neighbours add coherently; elsewhere they cancel. Real damping enters only through $(1 - i\epsilon)H$, p. 8.

1 Sign: $e^{-iH\delta t} - e^{-iA\delta t}e^{-iB\delta t} = +\frac{1}{2}[A,B]\,\delta t^2 + \ldots$, because the BCH exponents here are $-iA\delta t$ and $-iB\delta t$, and $(-i)^2 = -1$. The accumulated error $\sim t\,\delta t \to 0$ is unchanged.

For any small amplitude considered, using the Campbell – Baker – Hausdorf formula

$$\exp(A+B) = \exp A \exp B \exp(-\tfrac{1}{2}[A,B] + \cdots)$$

$$e^{-iH\delta t} = e^{-i\frac{p^2}{2m}\delta t}\, e^{-iV(q)\delta t}\, e^{[O(\delta t^2)]}$$

↑ additional terms if $[A,[A,B]] \neq 0$ or $[B,[A,B]] \neq 0$ (1)

The error I am making is

$$e^{-iH\delta t} - e^{-iA\delta t}e^{-iB\delta t} = -\frac{[A,B]}{2}\delta t^2 + \ldots \xrightarrow{\text{upon integration}} -\frac{[A,B]}{2}\delta t \xrightarrow{\text{further limit}} 0$$

for small δt ~~$\ll \hbar/\ldots$ & other scales of this expt~~

we approximately have

$$\langle q_2|e^{-iH\delta t}|q_1\rangle = \int dp_1 \langle q_2|e^{-i\frac{p^2}{2m}\delta t}|p_1\rangle\langle p_1|e^{-iV(q)\delta t}|q_1\rangle$$

$$= \int dp_1\, e^{-i\frac{p_1^2}{2m}\delta t}\, e^{-iV(q_1)\delta t}\, \langle q_2|p_1\rangle\langle p_1|q_1\rangle$$

$\langle q_2|p_1\rangle\langle p_1|q_1\rangle \sim \frac{1}{(2\pi)^{1/2}} e^{-ipq}$

$$= \int dp_1\, e^{-i\frac{p_1^2}{2m}\delta t}\, e^{-iV(q_1)\delta t}\, \frac{1}{(2\pi)}\, e^{-i\,p_1(q_1-q_2)}$$

$$= \int \frac{dp_1}{(2\pi)}\, e^{-iH(p_1,q_1)\delta t}\, e^{-ip_1(q_1-q_2)}$$

Trotter Product formulas:

$$\lim_{N\to\infty}(e^{-iA\varepsilon}e^{-iB\varepsilon})^N = e^{-i(A+B)t}$$

$t \equiv N\varepsilon$.

(4)

1 $\bar{q}$ is the average of the slice's two endpoints. For $P^2/2m + V(Q)$ the endpoint form of p. 3 has the same $\delta t \to 0$ limit; the midpoint matters when H mixes P and Q. For Weyl-ordered H, $\langle q_2|H|q_1\rangle = \int \frac{dp}{2\pi} H(p,\bar{q})\, e^{ip(q_2-q_1)}$ exactly.

Here $\mathcal{H}(P,Q)$ is simple and we could do separation of variables, but for more complex ones, e.g. $V(Q) \to V(P,Q)$. We can adopt

Wyle Ordering ← from classical hamiltonian $\mathcal{H}(p,q)$

$$\mathcal{H}(P,Q) \equiv \int \frac{dx}{2\pi}\frac{dk}{2\pi}\, e^{-ixP+ikQ} \int dp\,dq\, e^{-ixp-ikq}\, \mathcal{H}(p,q)$$

$\mathcal{F}$ $\mathcal{F}$

$\delta(P-p)$

Then the corrected transition amplitude would be (1)

$$\langle q_2|e^{-i\mathcal{H}\delta t}|q_1\rangle = \int \frac{dp_1}{2\pi}\, e^{-i\mathcal{H}(p_1,\frac{1}{2}(q_1+q_2))\delta t}\, e^{ip_1(q_2-q_1)}$$

Adopting Weyl ordering for the general case:

$$\langle q'', t''|q', t'\rangle = \int \prod_{k=1}^{N} dq_k \prod_{j=0}^{N} \frac{dp_j}{2\pi}\, e^{ip_j(q_{j+1}-q_j)}\, e^{-i\mathcal{H}(p_j,\bar{q}_j)\delta t}$$

isn't averaging over initial & final more satisfying?

$\bar{q}_j = \frac{1}{2}(q_j + q_{j+1})$, $q_0 = q'$, $q_{N+1} = q''$

If we now define $\dot{q}_j = \frac{q_{j+1}-q_j}{\delta t}$ and take the limit $\delta t \to 0$, we get

$$\langle q'', t''|q', t'\rangle = \int \mathcal{D}q\,\mathcal{D}p\, \exp\left[i\int_{t'}^{t''} dt\,(p(t)\dot{q}(t) - \mathcal{H}(p(t),q(t))\right] \quad \cdots (6.8).$$

One can further simplify by noticing that for simple enough $\mathcal{H}(p,q)$

(p less than or equal to p^2, and p^2 has no q dependence)

→ (this is Gaussian and the normalization / prefactor)

$$\langle q'', t'' | q', t' \rangle = \int \mathcal{D}q \exp\left[i \int_{t'}^{t''} dt\, L(\dot{q}(t), q(t)) \right] \quad (1)$$

The E.O.M is the replacement equation for p to finish the Gaussian integral.

$L(\dot{q}, q)$ is computed by first finding the stationary point of the p integral by solving

$$0 = \frac{\partial}{\partial p}(p\dot{q} - \mathcal{H}(p,q)) = \dot{q} - \frac{\partial \mathcal{H}(p,q)}{\partial p}$$

for p in terms of $\dot{q}$ and q, and then plugging back into $p\dot{q} - \mathcal{H}$. to get L.

1 Each p_j integral is Fresnel, not a real Gaussian: $\int \frac{dp}{2\pi} e^{i\delta t(p\dot{q} - p^2/2m)} = \left(\frac{m}{2\pi i\, \delta t}\right)^{1/2} e^{i\delta t\, m\dot{q}^2/2}$, the box continued to imaginary width; $(1 - i\epsilon)H$ (p. 8) makes it converge. The prefactor is q-independent and goes into the new $\mathcal{D}q$.

Gaussian integral

$$\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}. \longrightarrow \int_{-\infty}^{+\infty} e^{-(x+a)^2} d(x+a) = \sqrt{\pi}$$

$$\int_{-\infty}^{+\infty} e^{-|x/a|^2} dx = \frac{\sqrt{\pi}}{1/a}. \; a\sqrt{\pi}.$$

Quick derivation

$$I \equiv \int_{-\infty}^{+\infty} e^{-x^2} dx$$

$$I^2 = \int_{-\infty}^{+\infty} dx\, dy\, e^{-(x^2+y^2)}$$

$$= \int_0^{2\pi} d\theta\, dr \cdot r\, e^{-r^2}$$

$$= 2\pi \cdot \frac{1}{2} \int_0^{\infty} dr^2 e^{-r^2} = -\pi \left(e^{-r^2}\right)\Big|_0^{\infty} = \pi.$$

Now let's try to generalize:

$$\langle q'', t'' | Q(t_1) | q', t' \rangle = \langle q'' | e^{-iH(t''-t_1)} Q e^{-iH(t_1-t')} | q' \rangle$$

In the path-integral formula.
this $Q(t_1)$ will generate an extra $q(t_1)$

[e.g. from Eq 6.] $\langle q'', t'' | q', t' \rangle = \int \prod dq_k \prod \frac{dp_j}{2\pi} e^{i p_j (q_{j+1} - q_j) - iH\delta t}$ (Eq 6.)

$$\langle q'', t'' | Q(t_1) | q', t' \rangle = \int \mathcal{D}p\, \mathcal{D}q\; q(t_1) e^{iS}$$

where $S = \int_{t'}^{t''} dt\, (p\dot{q} - H)$

Now consider the opposite side $\int \mathcal{D}p \mathcal{D}q\; q(t_1) q(t_2) e^{iS}$

this clearly requires time ordering $Q(t_1)$ and $Q(t_2)$

if $t_1 < t_2$ $\quad Q(t_2) Q(t_1)$

if $t_1 > t_2$ $\quad Q(t_1) Q(t_2)$

so $\int \mathcal{D}p \mathcal{D}q\; q(t_1) q(t_2) e^{iS}$

$= \langle q'', t'' | \widetilde{T} Q(t_1) Q(t_2) | q', t' \rangle$.

↑ this is what appears in LSZ.

We can quickly realize that we are dealing with functionals.

We can definite the functional derivative / variation

$$\frac{\delta f(t_2)}{\delta f(t_1)} = \delta(t_2 - t_1)$$

(it satisfies the product rule, chain rule &.

Now, consider modifications to the Lagrangian.

$$H(p,q) \to H(p,q) - f(t)q(t) + h(t)p(t).$$

$f(t)$ and $h(t)$ are specified functions

$$\langle q'', t'' | q', t' \rangle_{f,h} = \int \mathcal{D}p\,\mathcal{D}q\; e^{-i\int_{t'}^{t''} dt\,(p\dot{q} - H + fq + hp)}$$

1 Each derivative needs its own $1/i$: $\frac{1}{i}\frac{\delta}{\delta f(t_1)}\frac{1}{i}\frac{\delta}{\delta f(t_2)}\langle\rangle_{f,h} = \int \mathcal{D}p\,\mathcal{D}q\, q(t_1)q(t_2)e^{iS}$. The same holds for every derivative in the last line of this page.

then

$$\frac{1}{i}\frac{\delta}{\delta f(t_1)}\langle q'', t'' | q', t' \rangle_{f,h} = \int \mathcal{D}p\,\mathcal{D}q\; q(t_1)\, e^{iS_{f,h}}.$$

$$\frac{1}{i}\frac{\delta}{\delta f(t_1)}\frac{\delta}{\delta f(t_2)}\langle\quad\rangle_{f,h} = \int \mathcal{D}p\,\mathcal{D}q\; q(t_1)q(t_2)\, e^{iS_{f,h}}$$ 1

$$\frac{1}{i}\frac{\delta}{\delta h(t_1)}\langle\quad\rangle_{f,h} = \int \mathcal{D}p\,\mathcal{D}q\; p(t_1) e^{iS_{f,h}}.$$

So we can get any # of p, q and then set

$h(t) = f(t) = 0$ that get us back to the original Lagrangian.

$$\langle q'', t'' | T(Q(t_1)\cdots P(t_n)\cdots | q', t' \rangle$$

$$= \frac{1}{i}\frac{\delta}{\delta f(t_1)}\cdots\frac{\delta}{\delta f(t_n)}\cdots\langle q'', t'' | q', t' \rangle_{f,h}\Big|_{f=h=0}$$

1 Take $t' \to -\infty$, $t'' \to +\infty$ at fixed $\epsilon > 0$, and $\epsilon \to 0^+$ only at the end. The projection uses eigenstates of the source-free H, so f, h must vanish at early and late times. (Above, $e^{iE_n t}$ means $e^{iE_n t'}$.)

(7)

If we are interested in initial and final states other than position eigenstates, we will need to integrate them over. We are also taking $t' \to -\infty$, $t'' \to +\infty$

$$\langle 0|0\rangle_{f,h} = \lim_{\substack{t'\to-\infty \\ t''\to+\infty}} \int dq''dq' \, \psi_0^*(q'') \langle q'', t''|q', t'\rangle \psi_0(q')$$

$$\psi_0(q) = \langle q|0\rangle, \quad \psi_0^* = \langle 0|q''\rangle$$

↑ ground state wave function in q-space

let $|n\rangle$ denote an eigenstate of $\hat{H}$ with eigenenergy E_n

suppose that $E_0 = 0$. If this is not true, we will shift H by such a constant.

$$|q', t'\rangle = e^{i\hat{H}t'}|q'\rangle = \sum_{n=0}^{\infty} e^{i\hat{H}t'}|n\rangle\langle n|q'\rangle$$

$$= \sum_{n=0}^{\infty} \psi_n^*(q') e^{iE_n t}|n\rangle$$

$\psi_n(q) = \langle q|n\rangle$ as the wave function of the nth eigenstate.

Now replace $\hat{H}$ with $(1-i\epsilon)H$, ϵ infinitesimal (positive) (1)

Then take the limit $t' \to -\infty$

Every state except the ground state is multiplied by a ~~fixed constant~~ exponentially vanishing factor of $\lim_{t\to-\infty} e^{t\epsilon E_n t}$.

2 A constant factor rather than a shift: $\int dq' \, \chi(q')\psi_0^*(q')$ and its q'' partner, nonzero if χ overlaps the ground state. Normalizing $\langle 0|0\rangle_{f=h=0} = 1$ removes it.

Similarly, this also picks out the ground state for $t' \to +\infty$

Futher, we can multiply arbitrary functions $\chi(q')$ and integrate over, which will yield a constant shift that can be absorbed into $\mathcal{D}q\mathcal{D}q$ [we then only need to care $\langle 0|0\rangle_{f,h}$] (2) Damping & project onto vacuum)

This implies that if we use

$H \to (1-\epsilon)H$, we can be [1] cavalier about boundary conditions ~~as~~.

Any reasonable boundary conditions will result in good ground state as both the initial & final state

$$\langle 0|0\rangle_{f,h} = \int \mathcal{D}p\mathcal{D}q\, e^{i\int_{-\infty}^{+\infty} dt\,(p\dot{q} - (1-\epsilon)H + fq + hp)}$$

Suppose $H = H_0 + H_1$

(note the connection with interaction picture, in contrast to Schrödinger Picture & Heisenberg Picture)

H_0 is solvable (e.g. free theory)

H_1 is a perturbation.

$$\langle 0|0\rangle_{f,h} = \int \mathcal{D}q\mathcal{D}q\, e^{i\int_{-\infty}^{\infty} dt\,(p\dot{q} - H_0(p,q) - H_1(p,q) + fq + hp)}$$

(H_0: $\to (1-\epsilon)$; H_1: $\to (1-\epsilon)$)

but we just for shorthand not write them out explicitly.

$$= e^{\left[-i\int_{-\infty}^{\infty} dt\, H_1\left(\frac{\delta}{i\delta h(t)}, \frac{\delta}{i\delta f(t)}\right)\right]} \times [2]$$

$$\int \mathcal{D}p\mathcal{D}q\, e^{i\int_{-\infty}^{\infty} dt\,(p\dot{q} - H_0(p,q) + fq + hp)}$$

we assume we know how to calculate this. (We shall see next a calculable example)

1 Read $(1-i\epsilon)H$ here and below, as on p. 8. $(1-\epsilon)H$ only rescales H and damps nothing; the $-i\epsilon$ is what gives $e^{-\epsilon E_n|t|}$.

2 The arguments need $1/i$: $H_1\left(\frac{1}{i}\frac{\delta}{\delta h(t)}, \frac{1}{i}\frac{\delta}{\delta f(t)}\right)$, since these bring down p and q (p. 7); cf. p. 10.

Now, if $\mathcal{H}_1$ only on q (not on p)
and if we (surely) are only interested in time-ordered
products of Q_s (not p_s), and if $\mathcal{H}$ is no
more than quadratic on P and the term quadratic in P
not dependent on Q, then we can simplify
the above equations to

$$\langle 0|0\rangle_f = e^{i\int_{-\infty}^{+\infty} dt\, L_1\left(\frac{1}{i}\frac{\delta}{\delta f(t)}\right)} \times$$

$$\int \mathcal{D}q \exp\left[i\int_{-\infty}^{+\infty} dt \left[L_0(\dot{q}, q) + fq\right]\right].$$

Here $L_1(q) = -\mathcal{H}_1(q)$.